



확산 모델 기반 모방학습 알고리즘을 이용한 자동 접안 시스템에 관한 연구

Seoul National University

Naval Architecture and Ocean Engineering

Jae Hong Min, Bo Woo Nam

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Introduction



Research Background

Implementations of autonomous docking systems have been widely studied using rule-based guidance^[1,2], potential fields^[3], and reinforcement learning^[4,5].

These methods usually assume the availability of precise global information about ships and obstacles, such as position and heading angle, which may not be available on ships without fine-grained sensing systems.

However, humans are capable of docking using only visual and inertial cues, which can be obtained from low-cost sensors such as RGB cameras and IMUs.

Research Objective

Development of an autonomous docking system through imitation learning to mimic human docking behavior.

[1] Martins Alfredo, et al. Autonomous surface vehicle docking manoeuvre with visual information. In ICRA, 2007.

[2] Yeongha Shin, et al. DOC* Algorithm: Docking Orientation Constrained Path Planning for Robust Autonomous Docking. IEEE Access, 2025.

[3] Woo Joohyun et al. Design of guidance law for docking of unmanned surface vehicle. Journal of ocean engineering and technology, 2016.

[4] Seo Ji Young et al. Auto-Docking for Leisure Boats Using Deep Reinforcement Learning With Action Masking. In OMAE, 2025.

[5] Pereira Maria Ines, et al. Reinforcement learning based robot navigation using illegal actions for autonomous docking of surface vehicles in unknown environments. Engineering Applications of Artificial Intelligence, 2024.

[6] Kim Moo Jin et al. Openvla: An open-source vision-language-action model. In arXiv, 2024.

[7] Black Kevin, et al. pi_0 : A Vision-Language-Action Flow Model for General Robot Control. In arXiv 2024.

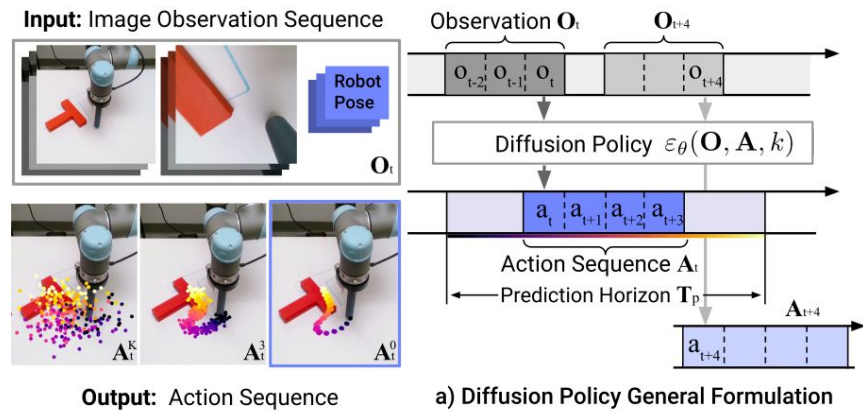
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Diffusion Policy^[1]

While diffusion models are usually used in generation tasks^[2,3,4],

it can also be used for a visuomotor policy.

In this case, diffusion models denoise action sequences, not images.



[1] Chi Cheng et al. Diffusion policy: Visuomotor policy learning via action diffusion. In RSS, 2023.

[2] Rombach Robin et al. High-resolution image synthesis with latent diffusion models. In CVPR, 2022.

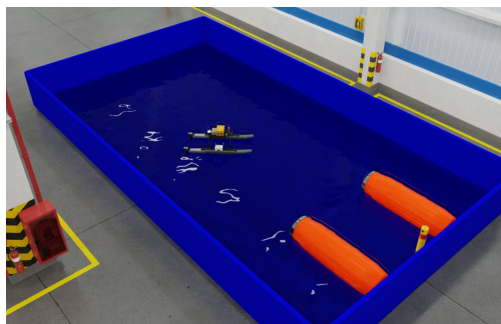
[3] Esser Patrick et al. Scaling rectified flow transformers for high-resolution image synthesis. In ICML, 2024.

[4] Jia Yueru et al. DesignEdit: Unify Spatial-Aware Image Editing via Training-free Inpainting with a Multi-Layered Latent Diffusion Framework. In AAAI, 2025.

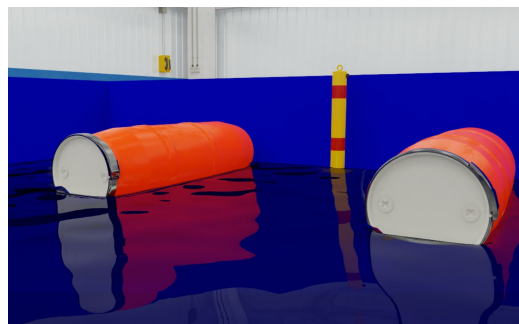
Simulation

IsaacLab Docking Simulator

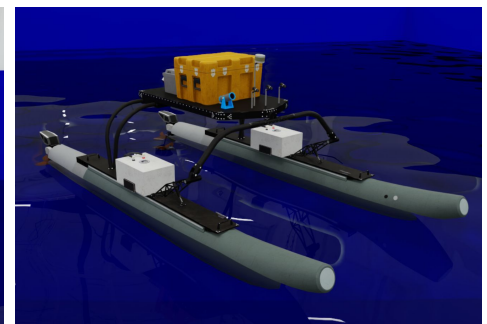
Built on IsaacLab^[1] using 3-DoF manoeuvring equation of motion^[2] following MarineGym^[3].



Overview



Berth



WAM-V from Virtual RobotX^[4]

[1] Mittal Mayank, et al. Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning. In arXiv, 2025.

[2] Fossen Thor I. Handbook of marine craft hydrodynamics and motion control. John Wiley & sons, 2011.

[3] Chu Shuguang, et al. MarineGym: a high-performance reinforcement learning platform for underwater robotics. In IROS, 2025.

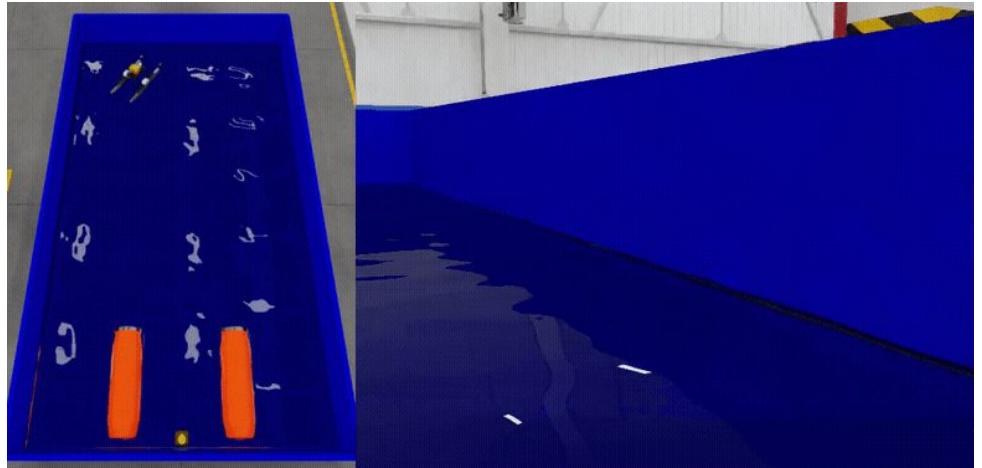
[4] Virtual RobotX (VRX) repository ([GitHub](#))

IsaacLab Docking Simulator

Collected demonstrations

Data Specifications

- Number of trajectories
 - 300
- RGB
 - Top-down view
 - Egocentric view
- IMU
 - Linear accelerations ($\ddot{x}, \ddot{y}$)
 - Angular velocities ($\dot{\psi}$)
- Motor Commands (15 Hz)



Top-down view

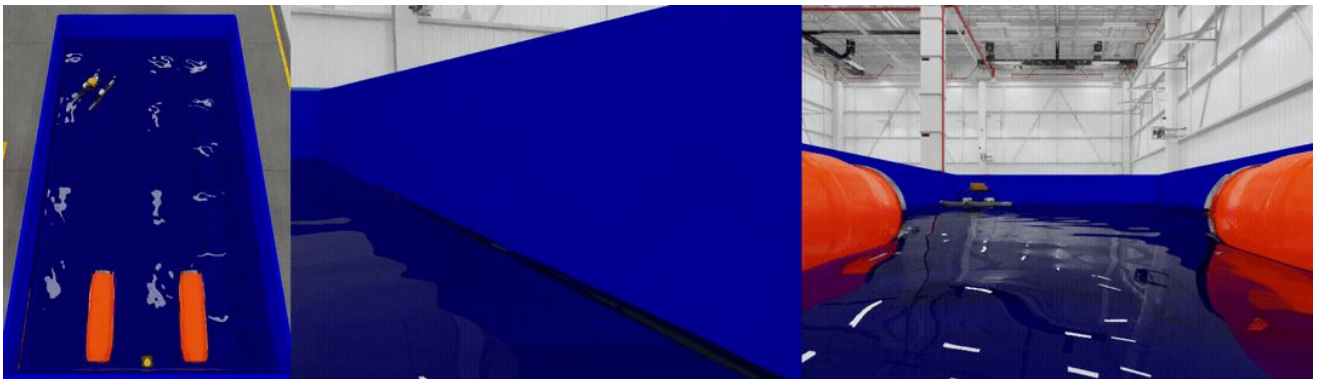
Egocentric view

[1] Mittal Mayank, et al. Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning. In arXiv, 2025.

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IsaacLab Docking Simulator

Success case



Successful rollout

[1] Mittal Mayank, et al. Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning. In arXiv, 2025.

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IsaacLab Docking Simulator

Failure case



Misalignment and contact

We did not observe any severe failure cases, such as direct collisions with the walls.

However, more precise docking remains as future work.

[1] Mittal Mayank, et al. Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning. In arXiv, 2025.

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Experiments

Test Pool Experiments

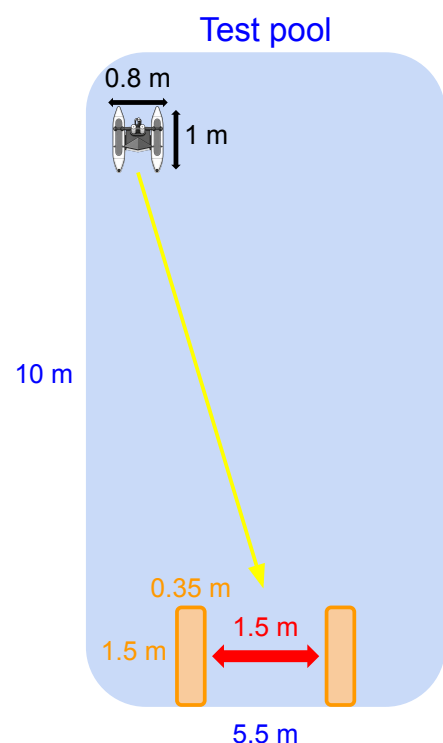
We collected real-world data in a test pool
and trained a visuomotor policy

Task: Head-on docking in calm water

Platform: A double-hull USV equipped with twin motors



Sample rollout of the trained policy



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Test Pool Experiments

Evaluation of the Trained Policy

Inputs

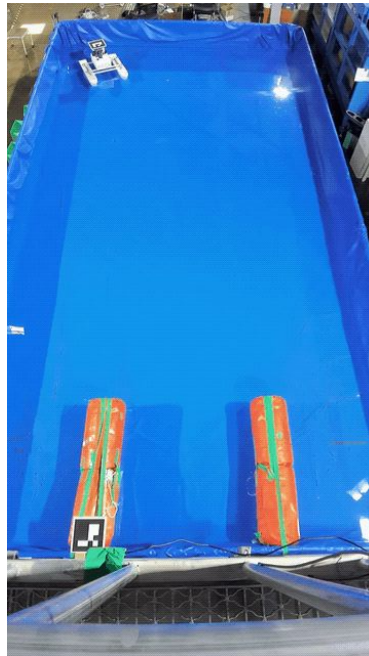
- Top-down, egocentric view images
- Linear accelerations, yaw rate

Success Criteria

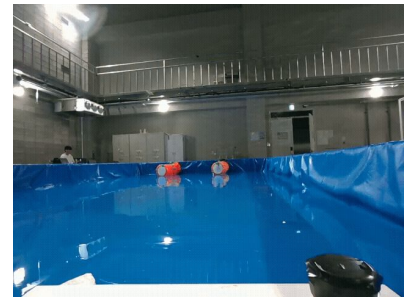
- Distance from the target point < 0.3 [m]
- Velocity of the ship < 0.1 [m/s]
- Heading error < 10 [°]

Success rate

- 90 % (Success: 18, Failure: 2)



Top-down view



Egocentric view



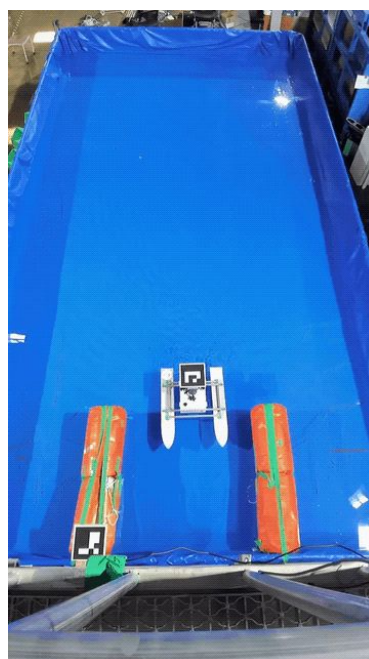
Aruco markers

Test Pool Experiments

Evaluation of the Trained Policy

Failure cases

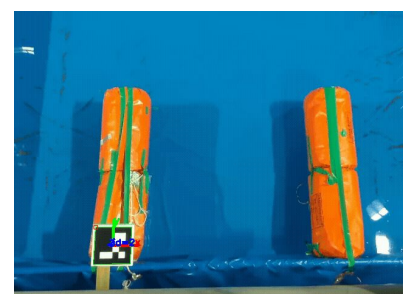
Insufficient deceleration led to contact with the pool walls.



Top-down view



Egocentric view



Aruco markers

Summary

Developed an autonomous docking system using imitation learning and evaluated policy performance in both simulation and real-world experiments.

Future Work

- Efficient Training
Reduce dataset size while maintaining performance
- Robust Docking under Disturbances
Handle wind, waves, and other environmental factors
- Cross-Embodiment
Transfer to different hull types and thruster configurations with little to no additional training
- Sim2Real Transfer
Pretrain in simulation and adapt to real-world with minimal fine-tuning data

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Thank you for your attention
